

PONTIAC, IL, USA

WMO: 722171

Lat: 40.924N

Lon: 88.625W

Elev: 659

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	2.8	-8.7	3.5	-0.7	-3.9	4.6	4.0	32.3	27.9	28.6	27.6	6.9	300	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	92.5	74.5	90.3	74.0	88.0	73.1	79.7	86.9	77.9	85.4	76.4	83.8	10.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	147.8	83.9	75.4	136.4	82.2	74.0	130.0	80.5	43.8	87.1	41.8	85.4	40.2	83.4	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.7	24.6	21.0	-11.1	95.5	8.2	3.0	-17.0	97.6	-21.8	99.3	-26.4	101.0	-32.3	103.1		
			-11.3	83.0	7.7	1.9	-16.9	84.4	-21.4	85.5	-25.8	86.5	-31.4	87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	26.0	28.1	40.8	52.0	63.8	72.6	74.3	72.3	67.3	54.2	41.8	31.0	(6)
(7)		DBStd	19.77	12.72	12.33	11.36	9.38	9.11	5.95	5.95	5.79	7.24	9.50	9.85	10.94	(7)
(8)		HDD50	2722	746	616	332	86	5	0	0	0	62	282	593		(8)
(9)		HDD65	5719	1209	1032	754	400	137	8	3	9	58	357	698	1054	(9)
(10)		CDD50	3504	2	4	48	146	434	677	752	691	519	193	35	3	(10)
(11)		CDD65	1023	0	0	4	9	100	235	290	235	127	23	0	0	(11)
(12)		CDH74	10294	0	0	40	165	1077	2292	2997	2173	1290	259	1	0	(12)
(13)		CDH80	3704	0	0	8	27	393	842	1155	746	463	70	0	0	(13)
(14)	Wind	WSAvg	9.2	11.6	11.2	11.1	12.1	10.3	7.9	5.4	4.6	6.0	9.2	10.6	10.7	(14)
(15)	Precipitation	PrecAvg	34.6	1.8	1.3	2.6	3.4	4.0	3.8	4.0	3.3	3.3	2.5	2.7	2.2	(15)
(16)		PrecMax	49.0	6.2	3.4	5.1	7.9	10.0	11.6	8.6	8.7	11.8	7.4	8.2	8.7	(16)
(17)		PrecMin	21.1	0.0	0.0	0.6	0.4	0.6	0.3	0.4	0.3	0.0	0.1	0.7	0.3	(17)
(18)		PrecStd	6.3	1.3	0.8	1.2	1.6	2.3	2.4	2.0	2.1	2.4	1.7	1.7	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	64.5	80.5	83.5	92.7	94.6	98.6	94.8	92.6	87.8	72.4	62.5	(19)
(20)		MCWB	56.7	58.5	64.6	63.1	69.7	74.4	77.6	75.1	73.0	71.0	59.2	58.8		(20)
(21)		2%	DB	53.8	57.0	70.4	79.1	88.4	91.0	91.4	90.6	89.8	81.1	66.1	55.3	(21)
(22)		MCWB	51.2	52.5	59.8	61.6	68.0	73.5	77.6	75.0	72.5	64.8	58.2	51.8		(22)
(23)		5%	DB	46.9	49.5	64.0	73.5	84.2	88.4	89.7	87.8	85.7	75.5	62.3	50.3	(23)
(24)		MCWB	44.8	45.9	56.9	59.4	66.5	72.6	76.6	74.2	70.6	63.4	55.9	47.7		(24)
(25)		10%	DB	42.4	43.5	58.8	69.8	80.8	84.4	86.2	83.9	81.8	70.6	57.0	45.8	(25)
(26)		MCWB	39.9	40.3	52.7	57.8	65.4	71.2	75.3	73.4	67.2	62.0	51.4	43.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	60.0	65.8	67.6	74.5	80.4	83.2	81.0	77.9	73.3	64.1	60.0	(27)
(28)		MCDWB	60.2	63.4	76.2	76.9	83.4	87.5	89.9	88.3	87.0	81.8	68.2	61.9		(28)
(29)		2%	WB	51.4	52.9	61.9	64.4	72.4	77.5	80.6	78.5	75.6	70.1	59.4	53.3	(29)
(30)		MCDWB	53.4	57.1	68.5	73.0	81.7	85.2	87.7	85.8	84.2	76.5	63.7	54.7		(30)
(31)		5%	WB	45.1	45.9	58.4	61.9	70.4	75.7	78.5	77.0	73.3	66.9	56.1	47.6	(31)
(32)		MCDWB	46.6	48.9	63.6	69.4	79.0	83.9	85.7	84.0	80.1	72.0	61.9	49.6		(32)
(33)		10%	WB	39.6	40.6	53.2	59.3	68.3	74.0	77.1	75.3	71.1	63.3	52.4	43.4	(33)
(34)		MCDWB	41.9	43.7	58.1	69.0	76.2	82.0	84.6	81.5	78.1	68.7	57.0	45.8		(34)
(35)	Mean Daily Temperature Range		MDBR	14.6	15.7	18.2	21.9	21.7	20.4	20.0	20.4	23.8	21.3	17.5	13.9	(35)
(36)		5% DB	MCDBR	19.3	23.4	24.8	29.2	26.7	23.3	22.0	23.9	26.6	27.5	24.0	19.1	(36)
(37)			MCWBR	16.7	18.3	16.4	15.1	11.6	10.1	10.1	10.8	11.0	14.0	16.3	16.3	(37)
(38)			MCDBR	18.6	21.8	22.7	24.0	21.5	19.3	18.8	18.7	21.3	20.8	21.1	18.0	(38)
(39)		5% WB	MCWBR	16.9	18.0	16.2	15.6	11.4	10.5	10.8	10.5	10.5	13.1	15.8	16.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.310	0.345	0.395	0.427	0.440	0.455	0.446	0.409	0.355	0.312	0.293	(40)	
(41)		taud	2.436	2.388	2.384	2.250	2.219	2.227	2.167	2.181	2.259	2.413	2.485	2.451	(41)	
(42)		Ebn at Noon	276	286	287	278	271	266	261	259	261	266	266	266	(42)	
(43)		Edh at Noon	25	30	34	42	44	44	46	44	38	30	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	550	836	1151	1493	1711	1947	1944	1713	1449	970	638	466	(44)	
(45)		RadStd	60	91	88	120	137	135	103	103	117	109	56	57	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page